



Intro

The Dropship Landing Pad in game (`Landing_Pad_Player`) has a sister version – the TPS Landing Pad. Visually the only difference is color, but to make the TPS Landing Pad work the same as the default version, we need to go into the Blueprints of the game.

My approach is to not alter the existing TPS Landing Pad in game – it's connected to quests after all. We could make it its own new object in the game separate from the original, but I want to try something else: Put both together into one item, but with two variations: `Landing_Pad_Player` and `Landing_Pad_Enemy`. We'll edit the DataTables (.jsons) and Blueprints to make a new version in the new color.

Programs

Icarus has its game data packed into .pak ("chunks"). The one used for light modding is simply called `Data.pak`, and it's automatically unpacked in IMM, which is how we access all the DataTables and their .json. But the game also has heavier chunks that are much bigger and have much more important files – that's where the assets and blueprints hide.

To access the files here, we must unpack the .pak files, which is done with UnrealPak in a command line..... Screw that, so we get programs to do it for us! Then once we find the files we want, we make a clone of the original Landing Pad and swap the textures for the other one.

Icarus Mod Manager by Jimk72

Vital part of modding, as you already know!

FModel

This is one of my favorite programs when modding, because it's nice and visual, and it does all the work for you – except the .uasset editing. It works as a sort of catalog for all the files in the game, and then you find the one you need, download it, and get to work.

(IMPORTANT NOTE: If you merge a file like normal in IMM, it won't automatically place the mod into the game's mod folder like normal, if FModel is open. Close the program when merging. Annoying, I know)



UAssetGUI

This is the main star of Blueprint editing (at least for me lol). Very user-unfriendly if you ask me, but for extremely simple swaps like these it won't cause too much trouble...

The program doesn't really need an installation; it simply sits as an .exe file wherever. When you open it for the first time, it'll ask which version we're using, so set it to 4.27.

Right-click on your .uasset files and click "Open with..." and choose UAssetGUI as the standard.

As you work in the program and make changes, it will spawn little backup copies of your files (.bak). Just delete them at the end of your editing so they don't make things cluttered.

DataTable

Firstly, we want to prepare the DataTables in IMM.

D_Deployable:

This DataTable defines which deployables exist under each object. So, we'll write in the name of our new variations under the first.

DEFAULT:
<pre>"Name": "Landing_Pad_Player", "Variants": [{ "RowName": "Landing_Pad_Player", "DataTableName": "D_DeployableSetup" }]</pre>
NEW:
<pre>"Name": "Landing_Pad_Player", "Variants": [{ "RowName": "Landing_Pad_Player", "DataTableName": "D_DeployableSetup" }, { "RowName": "Landing_Pad_Enemy", "DataTableName": "D_DeployableSetup" }]</pre>



D_DeployableSetup:

This is where we point to the deployable behavior for the item.

Here we copy the Enemy Pad, but under “DeployableBlueprint” we create our own to match the original – just with a new name (comes in a bit).

Neither `Landing_Pad_Player` nor `Landing_Pad_Enemy` have an icon here, which many other items have in this category, so we’ll insert their respective “DeployableIcon” here as well, so they will show up on the round variant picker. Leave the other stuff as default.

DEFAULT:
<pre>"Name": "Landing_Pad_Prop", "DeployableBlueprint": "/Game/BP/Objects/World/Items/Deployables/Prop/BP_Prop_LandingPad.BP_Prop_LandingPad_C"</pre>
NEW:
<pre>"Name": "Landing_Pad_Enemy", "DeployableBlueprint": "/Game/BP/Objects/World/Items/Deployables/LandingPad/BP_LandingPad_Enemy.BP_LandingPad_Enemy_C", "DeployableIcon": "/Game/Assets/2DArt/UI/Items/Item_Icons/Deployables/T_ITEM_Landing_Pad_Prop.T_ITEM_Landing_Pad_Prop"</pre>

Icarus Modding – Landing Pad

Documentation by Ferani



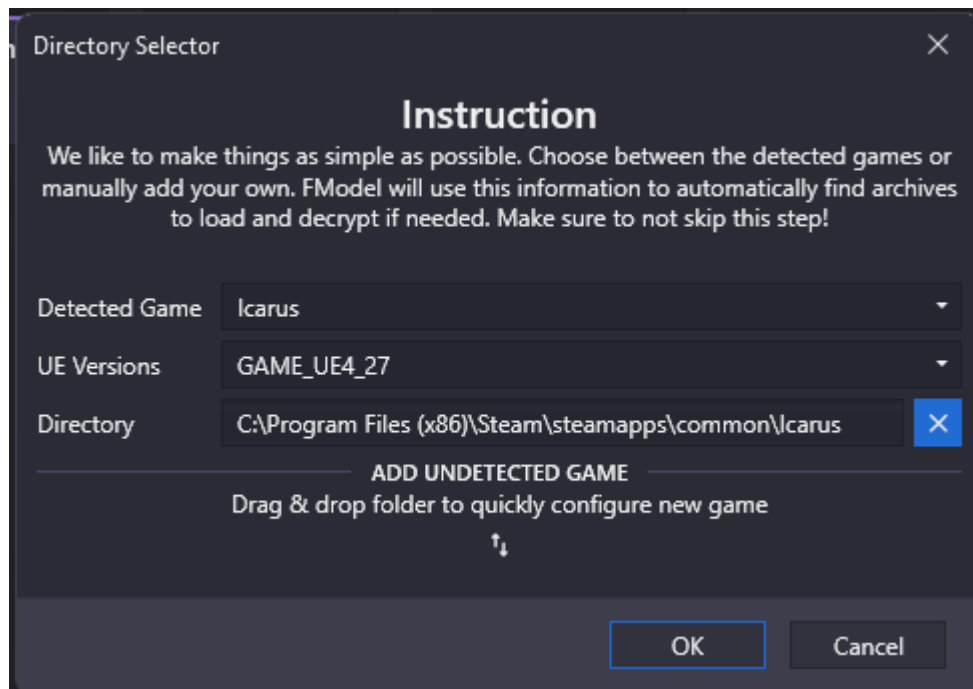
So here is our DT so far:

```
"version": "1.0",
"description": "The Dropship Landing Pad now has the TPS color variant.",
"fileName": "WIP_Landing_Pad",
"imageURL": "",
"readmeURL": "",
"Level1": "True",
"Rows": [
  {
    "CurrentFile": "Deployables-D_DeployableSetup.json",
    "File_Items": [
      {
        "Name": "Landing_Pad_Player",
        "DeployableBlueprint": "/Game/BP/Objects/World/Items/Deployables/LandingPad/BP_LandingPad_Player.BP_LandingPad_Player_C",
        "DeployableName": "NSLOCTEXT(\"D_DeployableSetup\", \"Landing_Pad_Player-DeployableName\", \"Player Landing Pad\")",
        "DeployableIcon": "/Game/Assets/2DArt/UI/Items/Item_Icons/Deployables/T_ITEM_Landing_Pad_Player.T_ITEM_Landing_Pad_Player",
        "PreviewStaticMesh": "/Game/ASS/DEP/LandingPad_T4/SM_DEP_LandingPad_T4_Combined.SM_DEP_LandingPad_T4_Combined",
        "DeployedSound": "/Game/FMOD/Events/SFX/D_Buildables/SFX_BLD_CONC.SFX_BLD_CONC",
        "SnapToSurfaceNormal": false,
        "MaxSurfaceSnapAngle": 20,
        "SupportsCustomRotation": false
      },
      {
        "Name": "Landing_Pad_Enemy",
        "DeployableBlueprint": "/Game/BP/Objects/World/Items/Deployables/LandingPad/BP_LandingPad_Enemy.BP_LandingPad_Enemy_C",
        "DeployableIcon": "/Game/Assets/2DArt/UI/Items/Item_Icons/Deployables/T_ITEM_Landing_Pad_Prop.T_ITEM_Landing_Pad_Prop",
        "PreviewStaticMesh": "/Game/ASS/DEP/LandingPad_T4/SM_DEP_LandingPad_T4_Combined.SM_DEP_LandingPad_T4_Combined",
        "DeployedSound": "/Game/FMOD/Events/SFX/D_Buildables/SFX_BLD_CONC.SFX_BLD_CONC",
        "SnapToSurfaceNormal": false,
        "MaxSurfaceSnapAngle": 20,
        "SupportsCustomRotation": false
      }
    ]
  },
  {
    "CurrentFile": "Traits-D_Deployable.json",
    "File_Items": [
      {
        "Name": "Landing_Pad_Player",
        "Variants": [
          {
            "RowName": "Landing_Pad_Player",
            "DataTableName": "D_DeployableSetup"
          },
          {
            "RowName": "Landing_Pad_Enemy",
            "DataTableName": "D_DeployableSetup"
          }
        ],
        "EffectedByWeather": false,
        "bMustBeOutside": true
      }
    ]
  }
]
```

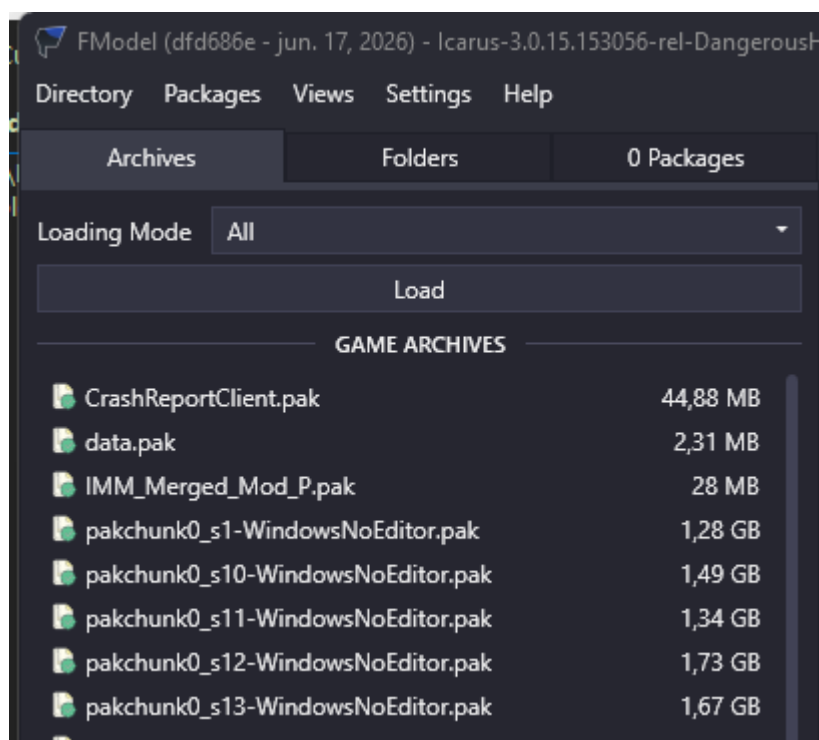


FModel

Time to get FModel! Download and install, and use the following directory setup:



On startup you will see a long list on the left called “Game Archives”. Set “Loading Mode” to “All” and press “Load”. This will open up all the chunks from the game and give us access to the files inside. Go to the folder “Icarus/Content” if you’re not already there by default.

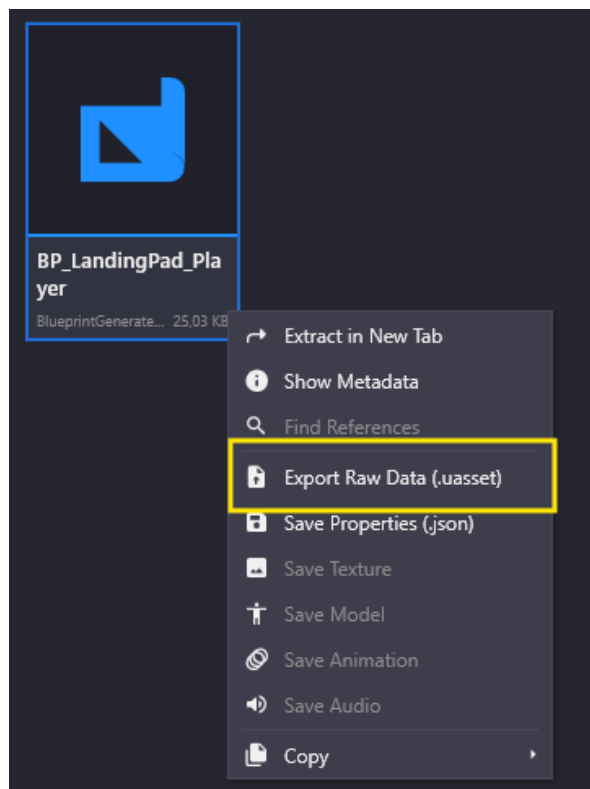




Browsing assets

We now need to find the BP's for the two landing pads, so we can compare and edit. You can search for them under Packages in the top bar of FModel, or you can go back to **D_DeployableSetup**, where it tells you where the “DeployableBlueprint” is.

- Icarus/Content/BP/Objects/World/Items/Deployables/LandingPad/**BP_LandingPad_Player**.uasset
- Icarus/Content/BP/Objects/World/Items/Deployables/Prop/**BP_Prop_LandingPad**.uasset



Locate them in FModel, right-click on the file, and choose “Export Raw Data (.uasset)”.

It'll show you at the bottom where the file ended up, and now move them to our mod folder that we made in IMM. .uasset files are always in pairs, and next to the BP_LandingPad_Player.uasset sits a BP_LandingPad_Player.uexp. We can safely ignore the .uexp, as long as we keep the pair together.

Now you need to copy the hierarchy that we found the original files in, but for our purposes we will place them both in the same folder, which is where our new DeployableBlueprint for **LPE** is. Keep its original name for now, and then we'll clone **LPP** in a bit.

Name	Date modified	Type	Size
BP_LandingPad_Energy.uasset	22-06-2026 10:14	UASSET File	26 KB
BP_LandingPad_Energy.uexp	22-06-2026 10:14	UEXP File	22 KB
BP_LandingPad_Player.uasset	21-06-2026 21:54	UASSET File	26 KB
BP_LandingPad_Player.uexp	21-06-2026 21:54	UEXP File	22 KB
BP_Prop_LandingPad.uasset	22-06-2026 10:20	UASSET File	10 KB
BP_Prop_LandingPad.uexp	22-06-2026 10:20	UEXP File	22 KB



UAssetGUI

Open both files side by side in UAssetGUI, and take a long, deep breath... Welcome to .uassets.

To clone `BP_LandingPad_Player`, click “File” and “Save as”, type in `BP_LandingPad_Enemy` and save it (don’t worry about the .uexp partner, it’s automatically generated). Now you should have three .uassets in your folder, open all three side by side:

- `BP_Prop_LandingPad.uasset`
- `BP_LandingPad_Player.uasset`
- `BP_LandingPad_Enemy.uasset`

You’ll be able to see that the original `LandingPad_Player` has a billion more folders (“exports”) than `Prop_LandingPad`. That is because `PLP` is primarily a decoration object, and because `LPP` has tons of code for its lights and landing pad functionality.

Open the “NameMap”, this is where all file names are stored, and thankfully it is connected to all its instances (if you edit any line here, any place in the file that uses that word gets changed, so you don’t have to go to try and find all instances yourself).

Since we just cloned the new `LPE` file and gave it a different name, we need to change it in UAssetGUI, so it knows its new name. Double-click on the lines in the file that are called anything with “`BP_LandingPad_Player`” and change them to the name for the `LPE` version and click Enter.

- `/Game/BP/Objects/World/Items/Deployables/LandingPad/BP_LandingPad_Player`
- `/Game/BP/Objects/World/Items/Deployables/LandingPad/BP_LandingPad_Enemy`
- `ExecuteUbergraph_BP_LandingPad_Player`
- `ExecuteUbergraph_BP_LandingPad_Enemy`
- `BP_LandingPad_Player_C`
- `BP_LandingPad_Enemy_C`
- `Default__BP_LandingPad_Player_C`
- `Default__BP_LandingPad_Enemy_C`

11	<code>/Game/BP/Objects/World/Items/Deployables/BP_DeployableBase</code>	utf-8	
▶ 12	<code>/Game/BP/Objects/World/Items/Deployables/LandingPad/BP_LandingPad_Enemy</code>	utf-8	
13	<code>/Game/BP/Objects/World/Items/Deployables/Lights/BP_Light_Electric_Base</code>	utf-8	

Ctrl+S to save, and before you close, go to View at the top bar and click “Recalculate Nodes”. This means that the directory gets updated with the new names if you make big changes to “Export” names, like editing a function name.



Change of plans

Now, this is where I looked inside and saw that things are a bit more complicated than I first assumed. Let's break it down:

The goal of cloning the **original object** for the **new version** is to simply swap out the meshes or materials that they use. For that we would find all the SM (Skeleton Mesh, the main mesh), SK (Skeleton, the bones of the mesh), M (Material, basically the skin) and T (Textures) for each object and do a clean swap between them, but the problem here is that they don't use the exact same files for an easy swap.

LPP			No mappings	4.27
	Name	Encod		
0	/Game/ASS/DEP/LandingPad_T4/ANI_DEP_LandingPad_T4_Lights_OpenClose	utf-8		
1	/Game/ASS/DEP/LandingPad_T4/M_DEP_LandingPad_T4_Lights_Off	utf-8		
2	/Game/ASS/DEP/LandingPad_T4/M_DEP_LandingPad_T4_Lights_On	utf-8		
3	/Game/ASS/DEP/LandingPad_T4/SK_DEP_LandingPad_T4_Lights	utf-8		
4	/Game/ASS/DEP/LandingPad_T4/SM_DEP_LandingPad_T4_Base	utf-8		
5	/Game/BP/AI/GOAP/EQS/EQS_FindNearestDropshipRelocation	utf-8		
6	/Game/BP/Building/Roads/BP_IcarusSplineConnectionComponent_Crude_Oil	utf-8		
7	/Game/BP/Building/Roads/BP_IcarusSplineConnectionComponent_Electric	utf-8		
8	/Game/BP/Building/Roads/BP_IcarusSplineConnectionComponent_Fuel	utf-8		

LPE			File	Edit	View	Utils	Help
	Name	Encod					
0	/Engine/BasicShapes/Plane	utf-8					
1	/Game/ASS/DEP/LandingPad_T4/ANI_DEP_LandingPad_T4_Lights_OpenClose	utf-8					
2	/Game/ASS/DEP/LandingPad_T4/M_DEP_LandingPad_T4_Concrete_Dark	utf-8					
3	/Game/ASS/DEP/LandingPad_T4/M_DEP_LandingPad_T4_Decals_A_TPS	utf-8					
4	/Game/ASS/DEP/LandingPad_T4/M_DEP_LandingPad_T4_Structure_Dark	utf-8					
5	/Game/ASS/DEP/LandingPad_T4/SK_DEP_LandingPad_T4_Lights	utf-8					
6	/Game/ASS/DEP/LandingPad_T4/SM_DEP_LandingPad_T4_Base	utf-8					
7	/Game/BP/Building/Roads/BP_IcarusSplineConnectionComponent_Crude_Oil	utf-8					
8	/Game/BP/Building/Roads/BP_IcarusSplineConnectionComponent_Electric	utf-8					

However, you can see that **LPP** has two extra Material files for lights, and **LPE** has two materials for structure, concrete, and decals. These won't make for a clean swap, we can't really see which files control materials, so we're gonna have to inspect the assets to see what we're working with.



Finding materials

Thankfully FModel can show us meshes in a little built-in viewer, so we don't have to export them and use another program to inspect them. You can double-click the files and see them in 3D, that's kinda fun.

Going to

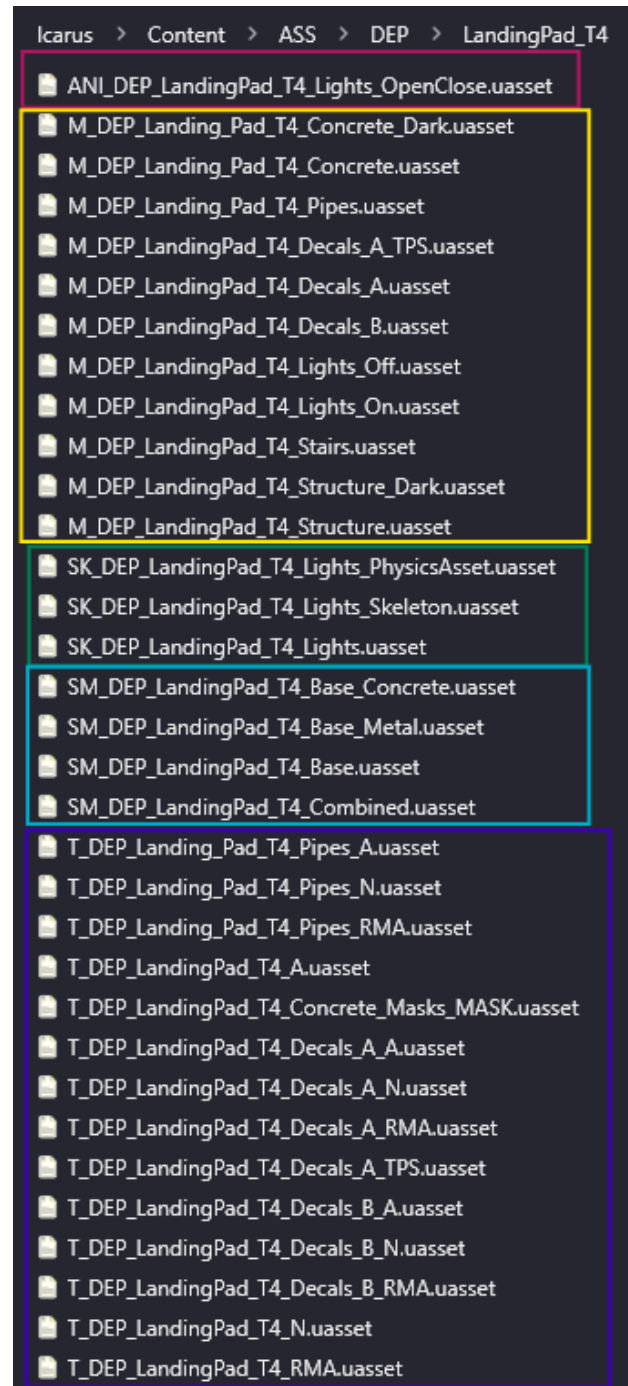
Icarus/Content/ASS/DEP/LandingPad_T4 gives us all the physical assets for the landing pads.

ANI and T are irrelevant for us, so we'll ignore those. The materials for the **LPE** are the versions ending in "Dark", so we'll keep those in mind.

We can recognize many of the files from the .uasset, but if they're all important, why don't we see all of them included in the file? That is because some of the assets are nested within each other:

For example, the .uasset only calls for **SK_DEP_LandingPad_T4_Lights.uasset**, but if we export that one and look at the NameMap, we can see that it includes the other files here, so it's one cohesive package.

That means that we can't do the simple swap in our **LPE** file alone, but if we clone the original assets that **LPP** uses and change the materials in those, we can then use those for **LPE**.





New Mesh-file

So going back to **LPP**, the file we want to clone and swap is

“SM_DEP_LandingPad_T4_Base.uasset”, so export its .uasset and put it into the folder next to the other files. Rename it (and the uexp) to “SM_DEP_LandingPad_T4_Enemy” and open it up.

If we look in the NameMap we can finally see the actual materials we want to swap!

Double-click on the lines in the file and change the names of the Concrete and Structure and click Enter.

For example:

- /Game/ASS/DEP/LandingPad_T4/M_DEP_Landing_Pad_T4_Concrete
- /Game/ASS/DEP/LandingPad_T4/M_DEP_Landing_Pad_T4_Concrete_Dark
- /Game/ASS/DEP/LandingPad_T4/M_DEP_LandingPad_T4_Structure
- /Game/ASS/DEP/LandingPad_T4/M_DEP_LandingPad_T4_Structure_Dark

Scroll down, and you can see the Material names pop up again, just without the path this time. Change Concrete and Structure to the dark versions here too. Save, Recalculate Nodes, and close.

Edit Blueprint

Now we’ve made the new SM ready with the new materials. Now we open our **BP_LandingPad_Enemy.uasset** again and go to the SM_DEP_LandingPad_T4_Base instances and change it to our new file. Don’t forget to change the path too, because we placed it next to our other working files, so it’s important the code knows where to find the new file.

From: /Game/ASS/DEP/LandingPad_T4/SM_DEP_LandingPad_T4_Base

To:

 /Game/**BP/Objects/World/Items/Deployables/LandingPad**/SM_DEP_LandingPad_T4_Enemy

Scroll down and find any instance of the old name and replace it with the new name. Save, Recalculate Nodes, and close.



In summary

We made two new .uasset files in our mods folder:

SM_DEP_LandingPad_T4_Enemy, which is a clone of the base mesh, except we swapped the materials for the darker versions.

BP_LandingPad_Enemy which is the DeployableBlueprint for the new variant of the landing pad, and the base-mesh was swapped for our custom mesh which holds the new materials.

We made the original landing pad have two variants in D_Deployable, and defined those variables in D_DeployableSetup.



WIP Landing Pad

Ver: 1.0 By: stnlepoulpe

The Dropship Landing Pad now has the TPS color variant.

Files:

json files:
D_DeployableSetup
D_Deployable

uasset files:
BP_LandingPad_Enemy
BP_LandingPad_Player
SM_DEP_LandingPad_T4_Enemy





Well... about that

I hoped that would be it, but unfortunately the new dark variant didn't have the same functionality, it couldn't be claimed by the player and thus wouldn't call for the ship to land on it.

The reason was that the game's claim interaction only accepts objects called "BP_LandingPad_Player". Our first black pad was a copy/sibling of the original landing pad, but it had a different name from the original, so the game did not recognize it as a valid landing pad.

We first looked into BP_DropShip, but that turned out to be a much more complicated file than we had thought (I think the game developers had a lot of big plans for orbit and dropships that were never implemented, so lots of noise in there), and it caused the game to get stuck on the loading screen.

Instead, we completely overhauled the BP_LandingPad_Enemy file and turned it into a "child" of BP_LandingPad_Player. Like how BP_Mount_Horse is a child of BP_Mount_Base, where Base has 90% of the functionality of what mounts do, and Horse only has the specific horse values and materials. That would mean that the game can call for LPP, but if LPE answers it's accepted.

Something-something about "SuperIndex" and "SuperStruct", we landed on an edit of LPE that made the child take on all of LPP's functionality, and only overrides the materials, which was the SM_DEP_LandingPad_T4_Enemy we made.

Then the landing pad worked in both colors, and mission accomplished!

Big shoutout to Codex for the diagnostics and overhauling of the file, because I would not otherwise have been able to make this possible... But I hope the documentation at least gives a peek into the process of modding, and how some of it is a relatively easy swap by hand if you have the right files, and sometimes you gotta basically rewrite parts of the game from scratch to get something to work how you want.